

AIO 2026 problems and solutions

a person

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Don't remember the problem names or description for p1. Just a placeholder from 1 sep until they get released on orac. Solutions are at end

Travelling

Time Limit: 1 second **Input File:** Standard Input
Memory Limit: 256 MB **Output File:** Standard Output

[I know this statement is wrong] You are travelling in a city with houses (in this order, also sorted) at points $A_1, A_2, \dots, A_n$. Find the maximum distance between 2 adjacent houses you travel to.

Input Format

The first line of input contains the integer n , the number of houses you will visit. The next line of input contains n space-separated integers, the ordered positions of houses.

Output Format

Output a single integer, the maximum distance between 2 adjacent houses you travel to.

Subtasks & Constraints

Forgot the subtasks if there were some, but $2 \leq N \leq 10^6$, $0 \leq A_i \leq 10^6$ for all i .

WHO

Time Limit: 1 second **Input File:** Standard Input
Memory Limit: 32 MB **Output File:** Standard Output

There are a bunch of delegates on a round table discussing important matters at the WHO.

A delegate can only speak after someone adjacent to them. Also, a delegate can only speak after someone less important than them. Determine if it is possible that all delegates can speak.

Input Format

The first line of input contains the integer n , the number of delegates. The next line of input contains n space-separated integers $A_1, A_2, \dots, A_n$, the importance of each delegate.

Output Format

You should output "YES" if it is possible, and "NO" otherwise (without quotes)

Subtasks & Constraints

For all subtasks, $1 \leq N \leq 10^6$, and $0 \leq A_i \leq 10^9$ for all i .

Subtask	Points	Additional Constraints
1	30	$N \leq 10^3$
2	70	No additional Constraints

Buying stuff

Time Limit: 1 second **Input File:** Standard Input
Memory Limit: 256 MB **Output File:** Standard Output

You are buying some stuff, but the government has made it better so that you will never spend more than K dollars per M days.
 Each day, you will try buy an item for A_i dollars. Businesses need to make money, so they will try to charge you the maximum amount under A_i possible, while conforming to government regulations.
 Find the total amount of money you will spend.

Input Format

The first line of input will contain 3 space-separated integers N, M, K , the number of days you will buy items, and M, K as above. The next line of input will contain N space-separated integers $A_1, A_2, \dots, A_n$, the prices of the items you want to buy.

Output Format

Output the total amount of money you will spend.

Subtasks & Constraints

For all subtasks, $N \geq 0$ and $0 \leq A_i \leq 10^9$ for all i .

Subtask	Points	Additional Constraints
1	30	$N \leq 10^3$
2	70	$N \leq 10^6$

Traveling II

Time Limit: 1 second **Input File:** Standard Input
Memory Limit: 256 MB **Output File:** Standard Output

You are traveling into a town where everyone is grumpy. So therefore you need to be quiet. At the start, your car is at volume 0. Then, for each kilometer in the town, you can play your music at a maximum volume of X_i . For each kilometer, you can either keep your volume the same, -1, or +1. Find the maximum sum of volume you can get.

Input Format

The first line of input is N , the length in kilometers of the town. The next line of input contains N space-separated integers $X_1, X_2, \dots, X_n$.

Output Format

Output the maximum sum of volumes you can get.

Subtasks & Constraints

For all subtasks, $N \geq 0$ and $0 \leq X_i \leq 10^4$ for all i .

Subtask	Points	Additional Constraints
1	20	$N \leq 10^3$, and for all $1 \leq i \leq N - 1$, $X_i = 10^4$
2	30	$N \leq 10^3$, and for all i , X_i is either 0 or 10^4
3	30	$N \leq 10^3$
4	20	$N \leq 2 \cdot 10^5$

Money

Time Limit: 1 second **Input File:** Standard Input
Memory Limit: 256 MB **Output File:** Standard Output

Everyone has a bunch of money, and the government wants to look good, so they want to maximise the money that the median person has. Each person has a bunch of money A_i , and you have k dollars to spare to give to people. What is the maximum median money that someone has (you cannot split dollars into smaller parts)?

Input Format

The first line of input contains 2 space-separated integers N and K , the number of people living in the place, and the number of dollars you can give people. The next line of input contains N integers, $A_1, A_2, \dots, A_n$.

Output Format

What is the maximum possible median of the money of all the people?

Subtasks & Constraints

For all subtasks, $N, K \geq 0$, and $0 \leq A_i \leq 10^9$

Subtask	Points	Additional Constraints
1	20	$K = 0$
2	30	For all i , $A_i = 0$
3	25	$K \leq 2 \cdot 10^5$
4	25	$K \leq 10^9$

Mundane squares

Time Limit: 1 second **Input File:** Standard Input
Memory Limit: 256 MB **Output File:** Standard Output

Magic squares are cool but mundane squares are mundane. You need to determine if a mundane square is possible to construct, and if so - construct it. You are given the row and column sums for every row, and every column. Determine if its possible, and construct one.

Input Format

The first line of input contains 2 integers, N and K , the size of the square (N by N), and the maximum value of any number in the square (values cannot be negative). The second line of input contains N integers, $R_1, R_2, \dots, R_n$, the sum of elements in each row. The third line of input contains N integers $C_1, C_2, \dots, C_n$, the sum of elements in each column.

Output Format

If such a mundane square is impossible to construct, output "No". Otherwise, output "Yes", and then a possible mundane square.

Subtasks & Constraints

For all subtasks, $N, K \geq 0$, and $0 \leq R_i, C_i \leq 10^9$. Also, a *easy* square is when $N, R_i, C_i \leq 100$.

Subtask	Points	Additional Constraints
1	10	$R_i = 0$ for all i , easy square
2	20	$K = 1$, easy square
3	20	$K = 10^9$, easy square
4	30	Easy square
5	10	$N \leq 10^3$

1 Travelling - Solution

I know this seems trivial but there was some sort of other observation to this problem, but just loop through the array and get the max of $A_{i+1} - A_i$

2 WHO - Solution

There are 2 ways I know how to do this one.

1. Notice that the delegate with the lowest importance must speak first. Then, go from their position to the end of the array, and then the start of the array to their position, checking for monotonicity
2. Notice that if such an arrangement is possible, there is only 1 position where $A_{i+1} - A_i$ is negative. Then just loop through once and check this

3 Buying stuff - Solution

This is quite a simple sliding window problem. To make it easy, add a buffer of M days backwards. Then, for each day, just keep track of $\text{cost}[i]$, the money you spend on day i .

4 Travelling II - Half solution

Apparently this is a well-known problem with a very simple greedy. In contest, i did subtask 2 and almost 1:

1. Notice that the optimal solution is like a pyramid, go up in value and then down (unless the last position can handle going up still)
2. You can break the journey into segments of continuous 10^4 s, separated by 0s. Then it's also like a pyramid, with triangular numbers.

The full solution requires some intuition, and is just go from front to back, and back to front, checking for each kilometer the maximum you can go :(

5 Money - Solution

In my opinion, easier (a lot) than p4. I'll go through the subtasks:

1. Find the median of the array, quite trivial
2. Notice that you only need to change the top elements (median and above), so floor divide
3. Since K is small, just greedily assign each dollar to the smallest person
4. Either vectorise the sub3 solution, or binary search on the minimum (didn't get this :() with an $O(N \log^2 N)$

6 Mundane Squares

I have no idea how to do this, except subtask 1 is extremely trivial. Think creatively for subtask 2 and 3 because bounds are so small, but after is very hard